

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

CS1B - Actuarial Statistics

(Paper B)

May 2025

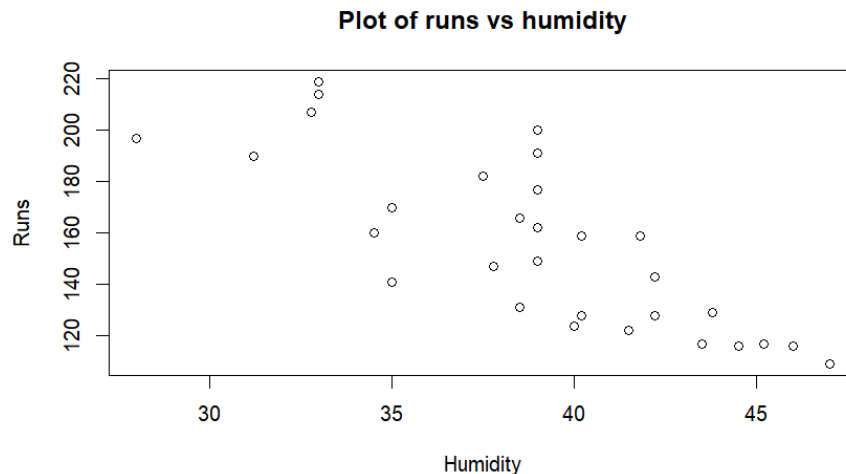
INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Q. 1)

- i) `runs<-read.csv("C:\\ runs.csv")` (1)
- ii) `plot(runs, main="Plot of runs vs humidity")`



(2)

- iii) `r<-cor(runs$Runs, runs$Humidity, method = "pearson")`
`r = -0.7942107`

There seems to a **strong negative** correlation between the humidity level and runs scored by the team.

(2)

- iv) `model_runs <- lm(Runs ~ Humidity, data = runs)`

`summary(model_runs)`

Call:

`lm(formula = Runs ~ Humidity, data = runs)`

Residuals:

Min	1Q	Median	3Q	Max
-37.05	-14.49	-1.82	14.95	44.73

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	377.4457	32.2818	11.692	2.74e-12 ***
Humidity	-5.6969	0.8237	-6.916	1.62e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 20.38 on 28 degrees of freedom

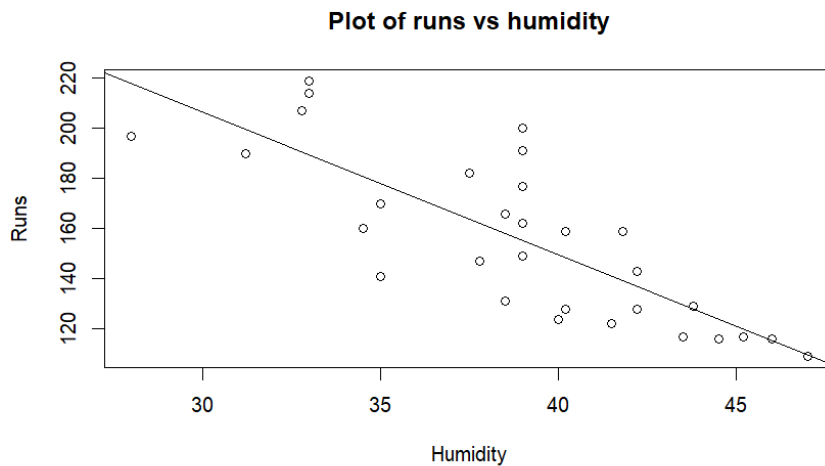
Multiple R-squared: 0.6308, Adjusted R-squared: 0.6176

F-statistic: 47.83 on 1 and 28 DF, p-value: 1.616e-07

The regression equation is $\text{Runs} = 377.45 - 5.6969 * \text{Humidity}$

(3)

v) `plot(runs, main="Plot of runs vs humidity")`
`abline(model_runs)`



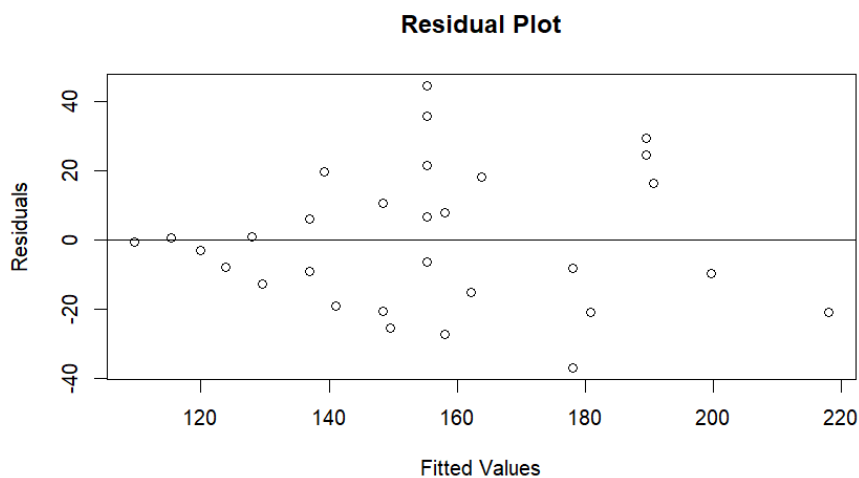
(2)

vi) `predict(model_runs, newdata = data.frame(Humidity = 48))`

1
103.9961

(2)

vii) `plot(model_runs$fitted.values, model_runs$residuals, xlab = "Fitted Values", ylab = "Residuals", main = "Residual Plot")`
`abline(h=0)`



The residuals are randomly scattered around 0 implying that linear model is appropriate

(3)

viii) `summary(model_runs)$r.squared`
`[1] 0.6307707`

$R^2 = 0.63$ means 63% of the variability in the dependent variable (runs) can be explained by the independent variable (humidity) from the model.

The model does a moderately good job at explaining how runs change with humidity.

(2)

ix) `confint(model_runs)`
 2.5 % 97.5 %
 (Intercept) 311.319524 443.571913
 Humidity -7.384142 -4.009593

CI for intercept is [311.31,443.57]

CI for slope is [-7.38,-4.01]

(2)

x) H_0 : No linear relationship between humidity and runs ($\beta = 0$) vs

H_1 : There is a linear relationship ($\beta \neq 0$)

```
anova_model <- anova(model_runs)
print(anova_model)
```

Analysis of Variance Table

Response: Runs

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Humidity	1	19862	19862.1	47.834	1.616e-07 ***
Residuals	28	11626	415.2		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

p-value = 1.616494e-07

Since p-value < 0.05 we reject the null hypothesis H_0 and

conclude that humidity has a statistically significant linear relationship with runs scored.

(6)

[25 Marks]

Q. 2)

i)

```
exercise_data <- matrix(c(38, 120, 42,
                          232, 357, 73,
                          48, 29, 4,
                          15, 34, 8),
                        nrow = 4, byrow = TRUE)
```

```
rownames(exercise_data) <- c("Sedentary", "Moderately Active", "Very Active",
                             "Extremely Active")
```

```
colnames(exercise_data) <- c("Light", "Moderate", "High")
```

```
print(exercise_data)
```

	Light	Moderate	High
Sedentary	38	120	42
Moderately Active	232	357	73
Very Active	48	29	4
Extremely Active	15	34	8

(3)

ii) H_0 : Exercise habit and workout intensity are **independent**

H_1 : Exercise habit and workout intensity are **not independent**

```
chisq.test(exercise_data)
```

Pearson's Chi-squared test

data: exercise_data

X-squared = 52.982, df = 6, p-value = 1.183e-09

p-value = 1.183e-09

Since p-value < 0.05, we reject H_0 and

conclude that exercise habits and workout intensity are not independent (5)

iii) before <- c(9.1, 8.0, 7.7, 10.0, 9.6, 7.9, 9.0, 7.1, 8.3, 9.6)
after <- c(8.2, 6.4, 6.6, 8.5, 8.0, 5.8, 7.8, 7.2, 6.7, 9.8)

mean(before)
[1] 8.63

mean(after)
[1] 7.5 (2)

iv) # Pearson
cor(before, after, method = "pearson")
[1] 0.7764615

Spearman
cor(before, after, method = "spearman")
[1] 0.8024353

Kendall
cor(before, after, method = "kendall")
[1] 0.6292532 (3)

v) Pearson's correlation coefficient measures the strength and direction of a **linear relationship** between the two sets of cholesterol levels, using the **actual numerical values** of the data.

Spearman's rank correlation and Kendall's Tau measure the strength of a monotonic relationship by using the **ranks** of the values rather than the raw data. These methods are **non-parametric** and are often more appropriate when the data is **ordinal** in nature. Hence, they are **more suitable here** (3)

vi) The mean cholesterol levels have **reduced** after following the workout routine (1)

vii) $H_0: \mu_{\text{before}} = \mu_{\text{after}}$ vs $H_1: \mu_{\text{before}} \neq \mu_{\text{after}}$

t.test(before, after, paired = TRUE, var.equal = TRUE)

Paired t-test

data: before and after

t = 4.7558, df = 9, p-value = 0.001036

alternative hypothesis: true mean difference is not equal to 0

95 percent confidence interval:

0.5925029 1.6674971

sample estimates:

mean difference

1.13

p-value = 0.001036

Since p-value < 0.05, we reject H_0 and

conclude that cholesterol level have changed after workout routine (5)

The health club claims that their workout routine has helped in reducing the cholesterol level of the members who enrolled for the program.

viii) $H_0: \mu_{\text{before}} = \mu_{\text{after}}$ vs $H_1: \mu_{\text{before}} > \mu_{\text{after}}$ (cholesterol reduced)

`t.test(before, after, paired = TRUE, alternative = "greater", conf.level = 0.99)`

Paired t-test

data: before and after

`t = 4.7558, df = 9, p-value = 0.0005179`

alternative hypothesis: true mean difference is greater than 0

99 percent confidence interval:

0.4596157 Inf

sample estimates:

mean difference

1.13

`p-value = 0.0005179`

Since `p-value < 0.01`, we reject H_0 and

conclude that cholesterol level have reduced after workout routine.

(5)

[27 Marks]

Q. 3)

i) `set.seed(2025)`

`X <- rbinom(n = 100, size = 1, prob = 0.4)` (2)

ii) `sum(X)`
43 (1)

iii) `set.seed(2025)`
`applications <- rbinom(n = 5000, size = 100, prob = 0.4)` (2)

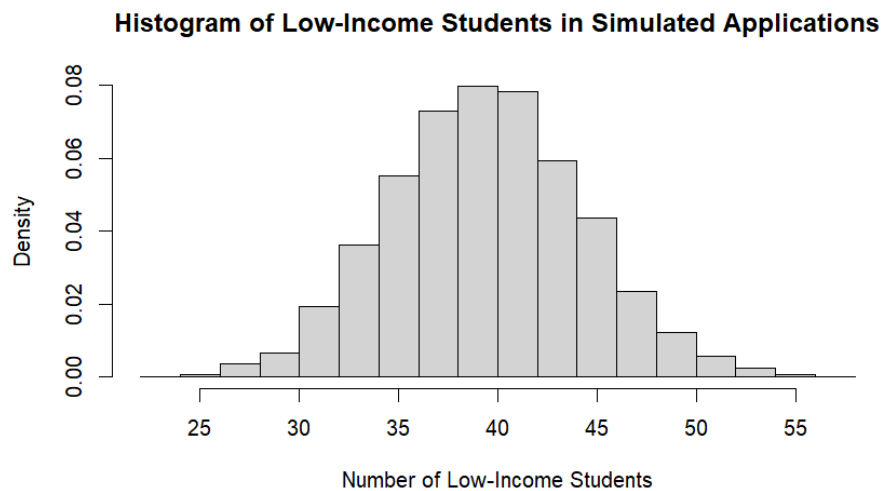
iv) `mean(applications)`
39.9536

`var(applications)`
23.7778 (2)

v) Theoretical mean $= n * p = 100 * 0.4 = 40$ (1)

vi) The simulated mean is very close to the theoretical mean which is expected since $n=5000$ simulations is large. (1)

vii) `hist(applications, main = "Histogram of Low-Income Students in Simulated Applications", xlab = "Number of Low-Income Students", probability = TRUE)`



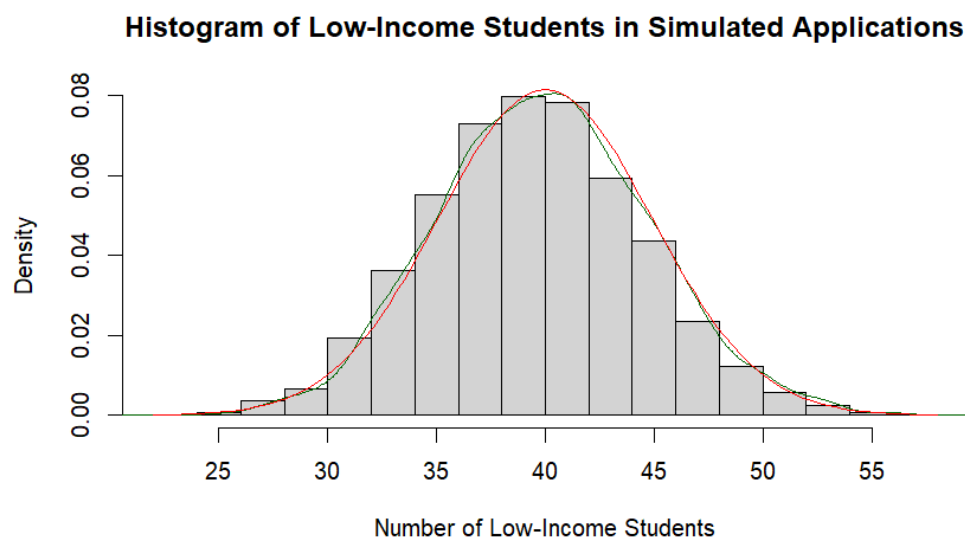
(2)

- viii) The sum of independent Bernoulli trials follows a Binomial distribution:
 $X \sim \text{Binomial}(n=100, p=0.4)$

Since n is large, this can be approximated by a Normal distribution with parameters np and $np(1-p)$ i.e
 $N(\mu=40, \sigma^2=24)$

(2)

- ix) `hist(applications, main = "Histogram of Low-Income Students in Simulated Applications",
 xlab = "Number of Low-Income Students", probability = TRUE)
 lines(density(applications), col = "darkgreen")
 curve(dnorm(x, mean = 40, sd = sqrt(24)), col = "red", add= TRUE)`



(3)

- x) The histogram of applications appears bell-shaped and symmetric, centered around 40.

The normal curve closely follows the shape of the histogram and density line.

This supports the central limit theorem that for large n , the $\text{Binomial}(n=100, p=0.4)$ distribution is well approximated by a Normal distribution.

(Max 2)
[18 Marks]

Q. 4)

i) `ev_data <- read.csv("C: \\ev_purchase.csv")` (1)

ii) `model_a <- glm(PurchaseEV ~ AvgDailyCommute + ChargingAccess + HouseholdAdults,
data = ev_data, family = binomial)`

`summary (model_a)`

Call:

`glm(formula = PurchaseEV ~ AvgDailyCommute + ChargingAccess +
HouseholdAdults, family = binomial, data = ev_data)`

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.6783	1.9802	0.343	0.732
AvgDailyCommute	-0.3497	0.3812	-0.917	0.359
ChargingAccess	0.1026	0.1644	0.624	0.533
HouseholdAdults	-0.1354	0.6004	-0.225	0.822

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 68.593 on 49 degrees of freedom
Residual deviance: 67.137 on 46 degrees of freedom
AIC: 75.137

Number of Fisher Scoring iterations: 4 (6)

iii) `> new_data <- data.frame(AvgDailyCommute = 3, ChargingAccess = 4, HouseholdAdults = 2)`

`predict(model_a, newdata = new_data, type = "response")`
0.4424472

As per the model there is a 44% chance that I will buy the EV. (3)

iv) `model_b <- glm(PurchaseEV ~ (AvgDailyCommute + ChargingAccess + HouseholdAdults)^2, data = ev_data, family = binomial)`

`summary (model_b)`

Call:

`glm(formula = PurchaseEV ~ (AvgDailyCommute + ChargingAccess +
HouseholdAdults)^2, family = binomial, data = ev_data)`

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-11.6367	11.6712	-0.997	0.319
AvgDailyCommute	2.0798	3.7019	0.562	0.574
ChargingAccess	2.4558	2.1868	1.123	0.261
HouseholdAdults	4.4745	3.9269	1.139	0.255
AvgDailyCommute:ChargingAccess	0.1851	0.2051	0.903	0.367
AvgDailyCommute:HouseholdAdults	-1.0317	1.2044	-0.857	0.392
ChargingAccess:HouseholdAdults	-0.9278	0.7178	-1.293	0.196

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 68.593 on 49 degrees of freedom
 Residual deviance: 63.173 on 43 degrees of freedom
 AIC: 77.173

Number of Fisher Scoring iterations: 5 (4)

- v) The p-values of interaction terms AvgDailyCommute:ChargingAccess, AvgDailyCommute:HouseholdAdults and ChargingAccess:HouseholdAdults are 0.367, 0.392 and 0.196 respectively.

All p-values are > 0.05 , thus they're not statistically significant. (2)

- vi) The AIC of model_b is 77.173 which is higher than the AIC of model_a (75.137). Thus, model_b is not an improvement over model_a (2)

- vii) `cost<-as.matrix(read.csv("C:\\ev_cost.csv"))`
`rownames(cost)<-c("A","B","C","D","E")`
`cost`

	Year_1	Year_2	Year_3	Year_4	Year_5	Year_6
A	9.8	10.5	11.0	12.3	15.2	18.5
B	21.5	22.0	23.3	25.0	22.5	20.1
C	6.0	6.5	7.2	8.7	9.8	10.5
D	12.2	13.0	14.3	16.5	19.0	22.0
E	7.9	8.5	9.5	11.0	12.8	14.5

(2)

- viii) `n<-ncol(cost)`
 6

`m<-mean(rowMeans(cost))`
 14.05333

`s<-mean(apply(cost,1,var))`
 7.6436

`v<- var(rowMeans(cost))-mean(apply(cost,1,var))/n`
 29.22376

`Z<-n/(n+s/v)`
 0.9582285 (8)

- ix) `P<- Z*rowMeans(cost)+(1-Z)*m`
 P

A	B	C	D	E
12.93221	22.05135	8.36465	16.07839	10.84007

(2)

[30 Marks]
